

Ankit Mukherjee, MSc

Computational Biologist

Barrackpore, Kolkata, West Bengal, India | +91 9330111538 | ankit0204.am@gmail.com

LinkedIn | GitHub

Research Interests

Computational Genomics • Population Genomics • Human Genetic Variation • Pharmacogenomics • Transcriptomics • Functional Genomics • Disease Biology • Precision Medicine • Statistical Genomics

Education

MSc in Bioinformatics (DBT-funded)

Savitribai Phule Pune University, Pune

2022–2024

CGPA: **9.93/10**

BSc in Microbiology

University of Calcutta, Kolkata

2019–2022

CGPA: **9.03/10**

Research Experience

Project Associate

2024–Present

CSIR–Institute of Genomics and Integrative Biology (CSIR–IGIB), Delhi

Population Pharmacogenomics and Precision Medicine – GenomeIndia Project

- Investigating the landscape of **clinically actionable pharmacogenomic variation** across the Indian population using whole-genome sequencing data from **10,000 individuals representing 83 genetically diverse Indian populations**.
- Curated and harmonized pharmacogenomic evidence from **PharmGKB, CPIC, PharmVar, and ClinPGx** to identify clinically actionable SNPs, indels, and star alleles relevant to drug response.
- Characterized **population-specific allele-frequency variation** across Indian populations and compared pharmacogenomic profiles with global populations to identify differences with potential implications for precision medicine.
- Analyzed clinically important pharmacogenes and star-allele haplotypes to infer **drug metabolizer phenotypes**, identifying population groups with potentially altered responses to commonly prescribed therapeutics.

GenomeIndia Database (GI-DB)

- Contributing author to **GI-DB**, a publicly accessible genomic database cataloguing genetic variation identified from **10,000 whole genomes representing 83 Indian populations**.
- Contributed to large-scale **variant processing, quality assessment, normalization, functional annotation and population-level analyses** supporting the construction of a comprehensive Indian genomic variation resource.

Database: ibdc.dbt.gov.in/gidb/

Blood Transcriptomics and Disease Progression in Friedreich Ataxia

- Investigating the **blood transcriptomic landscape of Friedreich Ataxia (FRDA)** to identify molecular signatures associated with disease state and clinical severity.
- Analyzing RNA-seq data from FRDA patients and controls to identify **differentially expressed genes and dysregulated biological pathways** associated with disease progression.
- Integrating transcriptomic profiles with clinical severity measures to investigate relationships between molecular changes and the **phenotypic heterogeneity of FRDA**.

iPSC-derived Cardiomyocyte Transcriptomics in Friedreich Ataxia

- Investigating the molecular basis of **FRDA-associated cardiomyopathy** using iPSC-derived cardiomyocyte models, addressing a major disease manifestation and cause of mortality in FRDA.
- Analyzing transcriptomic profiles across **Day 15 and Day 52 cardiomyocyte differentiation stages** to characterize transcriptional remodeling during cardiomyocyte maturation and disease progression.
- Comparing cardiomyopathic and non-cardiomyopathic FRDA models to identify molecular pathways associated with the transition from **FXN deficiency to cardiac pathology**.
- Investigating stage-specific transcriptional programs to identify candidate pathways associated with **cardiac dysfunction and disease heterogeneity**.

Research Mentorship

- Formally mentored a BTech student on the research project “**Haplotype Diversity and Shared Ancestry of**

- the **Sickle Cell Mutation (rs334) in Global Populations**”, providing guidance on population-genetic analysis, haplotype diversity, shared ancestry and interpretation of global allele-frequency patterns.
- Guided the student through computational analysis, interpretation of population-genetic results and scientific presentation of the research findings.

MSc Thesis

2023–2024

National Centre for Cell Science (NCCS), Pune

Computational Analysis of Viral Protein Evolution

- Investigated **positive selection in viral proteins** to identify amino-acid substitutions associated with evolution and potential functional adaptation.
- Developed custom amino-acid substitution matrices based on the **PAM framework** and mapped candidate substitutions onto protein structures to investigate their functional context.

Summer Research

2023

National Centre for Cell Science (NCCS), Pune

Population-associated SARS-CoV-2 Genetic Variation

- Investigated population-associated SARS-CoV-2 variants and their potential structural and functional consequences using sequence analysis and **molecular docking**.

Publications

1. Subramanian, K., Bhattacharyya, C., **Mukherjee, A***, et al. (2026). *An Atlas of Indian Genetic Diversity*. **medRxiv**. Preprint. DOI: 10.64898/2026.03.20.26348801
2. Mondal, D., Bhattacharyya, C., **Mukherjee, A.**, et al. (2026). *Capturing India's phenotypic diversity: Health insights from the GenomeIndia project*. **medRxiv**. Preprint. DOI: 10.64898/2026.04.01.26349926
3. Bhattacharyya, C., Subramanian, K., Uppili, B., **Mukherjee, A.**, et al. (2025). *Mapping Genetic Diversity with the GenomeIndia Project*. **Nature Genetics**, 57, 767–773. DOI: 10.1038/s41588-025-02153-x

*Equal first authorship. Ankit Mukherjee is bolded throughout.

Awards & Academic Distinctions

- **First Prize – Best e-Poster Presentation**, Indian Ataxia Conclave 2026, for research on transcriptomic signatures and disease progression in Friedreich Ataxia.
- **DBT-BET 2024 – Category II**, qualified with 214 marks.
- **All India Rank 51 – GAT-B 2022**, Graduate Aptitude Test in Biotechnology.
- **All India Rank 192 – IIT-JAM Biotechnology 2022**.
- **Qualified – TIFR Graduate School Examination 2022**, Biology stream.
- **Best Student Award – BSc Microbiology**, University of Calcutta.

Technical Expertise

Programming & Statistical Computing: Python (NumPy, Pandas) • R (Tidyverse, Bioconductor, ggplot2) • Bash/Shell (awk, sed) • Perl

Population & Medical Genomics: Population genomics • Pharmacogenomics • GWAS • Variant prioritization • Functional annotation

Transcriptomics & Functional Genomics: Bulk RNA-seq analysis • Single-cell RNA-seq analysis • Gene-set and pathway enrichment • Gene co-expression analysis (WGCNA)

Computational Infrastructure: HPC • SLURM • Git/GitHub • Conda/Bioconda • Docker • Singularity

Conferences & Scientific Presentations

- **Genomics India Conference 2026 – Accepted poster presentation:** “Landscape of Clinically Actionable Pharmacogenomic Variation Across Diverse Indian Populations from the GenomeIndia Project.”
- **Indian Ataxia Conclave 2026 – Presented research on blood transcriptomics and molecular signatures associated with Friedreich Ataxia disease severity; awarded First Prize for Best e-Poster Presentation.**

Languages

English: Professional proficiency **Hindi:** Professional proficiency **Bengali:** Native